WORKED SOLUTION

STEP I 2010, Question 11

Mechanics • Collision

STEP I 2010, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

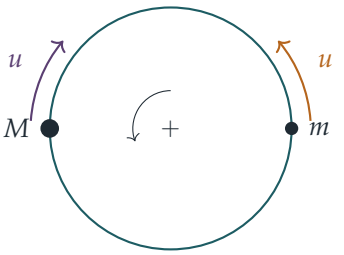
step.maths.org/questions/10-s1-q11

The key idea

Although the particles move on a circle, each collision is an ordinary one-dimensional collision along the common tangent: the smooth groove pushes only perpendicular to the motion, so it cannot change speeds or affect the tangential impulses. So we use conservation of momentum and Newton's experimental law with *signed* velocities measured round the circle. After that, two facts carry all the information: the total signed momentum never changes, and the relative speed is multiplied by e at every collision.

Solution

We model the particles as points in a smooth horizontal groove. Take anticlockwise as positive, with m initially moving anticlockwise at u and M clockwise at u, so their signed velocities are u and $-u$.



Between collisions the only horizontal force on each particle is the normal reaction of the smooth groove, which is perpendicular to the velocity. It does no work, so each particle keeps a constant speed. At a collision the impulses the particles exert on each other act along the common tangent, are equal and opposite, and the groove supplies no tangential impulse. So in each collision the total signed momentum round the circle is unchanged, and Newton's law applies to the signed velocities.

The first collision

Let the signed velocities after the first collision be p (for m) and q (for M). Conservation of momentum and Newton's law give

$$\begin{aligned} mp + Mq &= mu - Mu, \\ q - p &= e\big(u - (-u)\big) = 2eu. \end{aligned}$$

Substituting $q = p + 2eu$ into the first equation, $(m+M)p = (m-M)u - 2eMu$, so

$$p = -\frac{u\,(M - m + 2eM)}{M+m}, \qquad q = p + 2eu = \frac{u\,\big(2em - (M-m)\big)}{M+m}.$$

The speeds after the first collision are therefore

$$\text{speed of } m = \frac{u\,(M - m + 2eM)}{M+m}, \qquad \text{speed of } M = \frac{u\,|2em - (M-m)|}{M+m}.$$

Ask yourself Which signs mean that a particle has changed direction?

The particle m was moving in the positive sense, so it reverses exactly when $p < 0$. Since $M > m$ and $e \geq 0$, the bracket $M - m + 2eM$ is positive, so m *always* reverses. The particle M was moving in the negative sense, so it reverses exactly when $q > 0$, that is, when

$$2em > M - m.$$

So under this condition both particles change direction, as required.

After a further $2n$ collisions

Why $mv - MV = u(M-m)$. The total signed momentum is $mu - Mu = -u(M-m)$ at the start. Collisions conserve it, and between collisions the speeds don't change, so it keeps this value for ever.

Since both particles reverse at every collision, after the first collision m moves clockwise and M anticlockwise; each later collision swaps this round. After a further $2n$ collisions, $2n+1$ collisions have taken place in all, an odd number, so m is again moving clockwise and M anticlockwise. Their signed velocities are $-v$ and V, and

$$m(-v) + MV = -u(M-m) \quad \Longrightarrow \quad mv - MV = u(M-m).$$

The relative speed. After each collision the particles move apart in opposite directions round the circle with constant speeds, and they next meet on the far side, approaching each other at the same relative speed with which they separated. Newton's law then says that the speed of separation after a collision is e times the speed of approach before it. The initial speed of approach is $2u$, so after $2n+1$ collisions the speed of separation is $2ue^{2n+1}$. As the particles are moving in opposite directions, this is $v + V$:

$$v + V = 2ue^{2n+1}.$$

Solving. Multiply the second equation by M and add it to the first:

$$(m+M)v = u(M-m) + 2Mue^{2n+1},$$

and multiply the second by m and subtract the first:

$$(m+M)V = 2mue^{2n+1} - u(M-m).$$

Hence

$$v = \frac{u\big(M-m+2Me^{2n+1}\big)}{M+m}, \qquad V = \frac{u\big(2me^{2n+1}-(M-m)\big)}{M+m}.$$

Checks

- Putting $n = 0$ gives back the speeds after the first collision found above.
- With $m = 2$, $M = 3$, $e = 0.8$, $u = 1$, the formulae give $(v, V) = (1.16, 0.44)$ for $n = 0$ and $(0.8144, 0.2096)$ for $n = 1$. A step-by-step simulation of the particles going round the circle and colliding gives exactly these values after the 1st and 3rd collisions.
- Kinetic energy should not increase. With $e = 1$, $v = u\,\frac{3M-m}{M+m}$ and $V = u\,\frac{3m-M}{M+m}$, and you can check that $\frac{1}{2}mv^2 + \frac{1}{2}MV^2 = \frac{1}{2}(m+M)u^2$.

Common slips

- Mixing speeds and velocities: Newton's law and conservation of momentum must be written with signed velocities. Only at the end do you convert to speeds.
- Forgetting that $2n$ further collisions means $2n+1$ in total, which gives the power e^{2n+1} rather than e^{2n}.
- Worrying about the centripetal force from the groove. It is perpendicular to the motion, so it neither changes the speeds nor enters the collision equations.

Beyond STEP

The formula for V shows that the given assumption can't last for ever when $e < 1$: once $2me^{2n+1}$ falls below $M-m$, the heavier particle stops reversing. Momentum is conserved here only in a rotational sense. The groove's force always points through the centre, so the conserved quantity is really the *angular momentum* about the centre, and the circle's radius cancels. Conservation laws that come from symmetries like this are a central theme of first-year mechanics; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats angular momentum and collisions carefully.

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